

Ioannis-Aris Koufopoulos

MASTER'S GRADUATE · DATA SCIENCE

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Education

University of Piraeus

Piraeus

MSC IN CYBERSECURITY AND DATA SCIENCE

Oct 2022 - Jul 2024

- Specialization: Data Science
- Thesis: Fact-Checking with LLMs [PDF]
- Advisor: Dr. George Papastefanatos (Athena RC)

Athens University of Economics and Business

Athens

BSc IN INFORMATICS

Oct 2017 - Sep 2022

- Specializations: Computer Systems and Networks - Information systems and security

Professional experience

Research associate

Oct 2024-Present

ATHENA RESEARCH CENTER

- Built and optimized Retrieval-Augmented Generation (RAG) pipelines, including data ingestion, embeddings, vector search, and retrieval optimization.
- Constructed knowledge graphs from unstructured text using entity extraction and relationship modeling.
- Applied large language models (LLMs) for question answering, summarization, and text generation tasks.
- Developed machine learning and NLP models for text classification and semantic understanding.
- Collected, cleaned, and processed large-scale web text data to support RAG systems and model training.

Publications

Georgios Kaoukis, **Ioannis-Aris Koufopoulos**, Eleni Psaroudaki, Danae Pla Karidi, Evaggelia Pitoura, George Papastefanatos, and Panayiotis Tsaparas. Emeraldmind: A knowledge graph-augmented framework for greenwashing detection. In Proceedings of the ACM Web Conference 2026 (WWW '26), page 11, New York, NY, USA, 2026. ACM. 7 [PDF] [Code]

Georgios Kaoukis, **Ioannis-Aris Koufopoulos**, Eleni Psaroudaki, Danae Pla Karidi, Evaggelia Pitoura, George Papastefanatos and Panayiotis Tsaparas. EmeraldApp: An AI-Driven Tool for Detecting Greenwashing in Sustainability Claims. The ACM Web Conference 2026 (WWW '26), The 6th International Workshop on Computational Methods for Online Discourse Analysis (BeyondFacts '26). [PDF] [Code]

Skills

Programming Languages: Python, R, SQL, Java.

Tools and Frameworks: PyTorch, Hugging Face Transformers, spaCy, NLTK, scikit-learn, LangChain, Pandas, NumPy.

Specializations: Machine Learning, Natural Language Processing (NLP), Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), Knowledge Graph Construction, LLM Evaluation and Prompt-Based Reasoning, Data Harvesting and Cleaning.

Key Projects

Fact Checking Tool: Built an end-to-end Retrieval-Augmented Generation (RAG) tool for tracking and evaluating false claims in multiple languages (Greek and English).

Text summarization tool: Developed a text summarization tool that converts large corpora into concise bulleted summaries using large language models (LLMs).

Synthetic dataset generation: Generated a synthetic dataset for greenwashing detection, with the utilization of LLMs and the extraction of news articles.

Awards, Fellowships, & Grants

2024 **MSc Academic Achievement Award**, University of Piraeus

2024 **Merit-Based Scholarship**, University of Piraeus

Languages

English: C2 Proficiency – ECPE Certificate (University of Michigan).

Greek: Native